

USENA FLOW

Full System Architecture — Jepson Myles Studio

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Overview

USENA Flow is a full-stack photography studio workflow management platform built for Jepson Myles Studio. It manages the complete lifecycle of a photo project — from client booking to final delivery — across an entire team with multiple roles. The system integrates Google Calendar, Google Drive, Gmail, AI-powered features, and automated email workflows to streamline operations.

Tech Stack

Layer	Technology
Frontend	React 18 + TypeScript + Vite
UI Components	shadcn/ui (Radix UI) + Tailwind CSS
Routing	Wouter
Server State	TanStack Query v5
Forms	React Hook Form + Zod
Backend	Node.js + Express.js (TypeScript)
Database	PostgreSQL (Neon serverless) via Drizzle ORM
Validation	Zod schemas (shared between front and back)
Real-time	Server-Sent Events (SSE)
Auth	Role-based session auth + token-based client auth

User Roles

Role	Description
Admin	Full access — manages users, settings, templates, automations
Sales	Manages client communication, approves gallery deliveries
Lead Retoucher	Team lead, project assignment, quality oversight
Retoucher	Core editing workforce, rollover tracking, photo uploads
Data Wrangler	Handles shoot data, selects, project creation
Finance	Invoice and commission tracking
Client	Token-based access to their own chat only (no login)

Database Tables

Table	Purpose
users	Staff accounts with roles and credentials
projects	Core project records with 95+ fields covering every lifecycle stage
shoottracker_meta	1:1 with projects — calendar sync config, turnaround, holidays
project_notes	Text and image notes attached to projects
project_events	Audit trail of rollovers, completions, assignments
trade_offers	Retouchers swapping projects with each other
wrangler_commissions	1.5% commission tracking per wrangler on extras
complaints	Retoucher issue reports with image attachments
client_messages	WhatsApp-style chat messages per project
client_auth_tokens	Time-limited tokens for client chat access
chat_encryption_keys	End-to-end encryption keys per project chat
email_logs	Full log of every email sent through the system
calendar_events_staging	Google Calendar events pending promotion to projects
app_settings	Key-value config store (schedules, toggles, templates)
ai_memories	Persistent AI learning memories across interactions
ai_directives	Admin instructions injected into AI prompts
leave_requests	Staff leave requests with AI-powered approval

Table	Purpose
rewards / referrals	Client loyalty tiers and referral tracking
speed_records	Timestamps at each status transition for turnaround analytics

Backend Services

Service	Purpose
emailService	All branded emails via Resend — delivery, extras, surveys, referrals, delay notices, chat links
aiService	OpenAI integration — chat replies, coaching, risk prediction, workload forecast, quality reviews
aiMemoryService	Stores and retrieves AI learning memories, scored by importance
dataLearningService	Scans all data sources and extracts AI observations on demand
automationRegistry	Central registry of all automations — enable/disable toggles, activity logs
shoottrackerEngine	Calendar event parsing, project creation, due date calculation
calendarSync	Google Calendar polling and event staging
googleCalendar	Google Calendar API wrapper
googleDriveService	Creates client folders, monitors uploads, generates gallery links
driveMonitorService	Background watcher — detects when photos land in Drive, triggers delivery
gmailService	Gmail API wrapper for sending and reading email threads
gmailReplyMonitor	Monitors Gmail for client replies and routes them into chat
emailQueue	Queued email processing to prevent rate limit issues
chatAutoResponder	Sends automated comfort messages when team is offline
rewardsEngine	Calculates VIP tiers, bonus photos, referral rewards per client
riskCalculator	Scores each project as SAFE / AT_RISK / OVERDUE
delayNotificationScheduler	Sends scheduled delay alert emails to clients
morningCheckScheduler	Daily scan — risk updates, upcoming deadline alerts
defaultEmailTemplates	All base email templates with variable placeholders
icsCalendar	Generates ICS calendar files for shoot date confirmations

Background Schedulers

Scheduler	Trigger
autoSyncScheduler	Polls Google Calendar on a configurable interval
rolloverScheduler	Handles daily project rollover (currently manual-only mode)
morningCheckScheduler	Runs once daily at start of business
delayNotificationScheduler	Runs on a set cadence to send delay warnings
driveMonitorService	Continuously polls Drive folders for new uploads
gmailReplyMonitor	Polls Gmail for client email replies

Frontend Pages

Route	Purpose
/	Main dashboard — project cards, calendar, analytics, widgets
/editor-chat	Internal team-facing chat view per project
/client-chat	Token-authenticated client chat (PWA)
/ai-team-chat	AI-powered team Q&A chat (admin + retouchers)
/ai-brain	Admin panel for AI directives and memories
/automations	Visual hub for all system automations with logs
/shoottracker-settings	Google Calendar sync, turnaround config, holiday setup
/leave-management	Staff leave requests with AI workload-based approval
/approve-extras	Token-gated page for clients to approve extra charges
/survey	Post-delivery satisfaction survey
/referral	Client referral reward landing page

Key System Flows

Project Lifecycle

New booking !' ShootTracker syncs from Google Calendar !' staged as event !' promoted to project !' assigned to retoucher !' photos uploaded to Drive !' Drive monitor detects completion !' Sales approves gallery !' branded delivery email sent to client !' satisfaction survey triggered !' rewards calculated.

Client Communication

Project created !' chat link generated !' client gets email with secure token link !' client and editor exchange messages in real-time (SSE) !' end-to-end encrypted !' if team offline, auto-responder sends comfort message !' Gmail replies also routed into chat.

AI Layer

Every AI action reads active directives (admin rules) + relevant memories (past learnings) before generating a response. After each interaction, new observations are extracted and saved back to memory, scoring importance 1–10. Low-importance memories are pruned automatically.

Rewards Engine

Client project count !' VIP tier (Bronze/Silver/Gold/Platinum) !' bonus photos unlocked !' referral codes issued !' referral bookings tracked !' commission credited to wrangler.

External Integrations

Integration	Purpose
Google Calendar	Shoot booking sync !' project auto-creation
Google Drive	Photo delivery folder creation and monitoring
Gmail	Branded email sending + reply monitoring
Resend	Transactional email delivery
OpenAI	All AI features (GPT-4 vision + text)
Object Storage	Chat attachments, sneak peek photos, note images